Q-TRACE · October 13, 2026 · Implementation progress

One-page summary · meeting 3 of 7 · reports Phases 1–3B, completed October 6–7, 2026

Chronology. These milestones were completed on Oct 6–7, 2026. The full study, including the one-time sealed test, was completed on Oct 8. Nothing is still running; later meetings present the remaining results in depth.

Purpose. To show how the project went from raw data to a working, frozen quantum classifier, and what the first noise characterization found.

Key messages and numbers.

Milestone	Result	Source
Data (Phase 1)	284,807 → 283,726 rows after deduplication; 473 frauds (0.17%); 5 group-aware partitions; 0 groups cross partitions	`ulb_v1_*`
Baselines (Phase 2)	AUPRC 0.805 (random forest, 30 features); 0.59–0.62 on the VQC's 4 inputs	`phase2_classical_metrics.csv`, `p3ar_validation_comparison.csv`
Frozen VQC (3A-R)	4 qubits; ZZ feature map + RealAmplitudes (2 layers); parity readout; 12 parameters; 18 CX	`configs/repair/p3ar_v1.yaml`, `p5a_pool_metrics.csv`
Encoding repair	AUPRC 0.104 → 0.507 (AUROC 0.957); distinct fraud inputs 91 → 212 of 213	`p3ar_*`
First noise result (3B)	ΔAUPRC $\leq$ 0.003; Brier 0.0093 → 0.121; precision 0.53 → 0.03 (combined-high)	`p3b_exact_metrics.csv`

Takeaways.

- The data pipeline is leakage-safe.
- Classical models set the bar, so there is no quantum-advantage claim.
- A pre-registered, training-only repair made the VQC work.
- Noise damages calibration, not discrimination.

Slide	Content	Speaker
1	Title and chronology	Mayank
2	Leakage-safe data and baselines	Mayank
3	The frozen quantum classifier	Mayank
4	Encoding repair	Tushar
5	First noise result and project status	Tushar

Speaker split. Mayank takes the first half and Tushar the second, the same pattern as the committed Oct 13 script. The team should confirm it.

Related file. An earlier 8-slide whole-study version, `deliverables/presentations/Q-TRACE_Oct13_Progress.pptx`, is kept unchanged. Choose one deck for the meeting.

Needs confirmation.

- The meeting date, slot length and upload format.
- Whether the 5-slide milestone version or the earlier 8-slide whole-study version fits the session.